TUTORIAL NOTES

Quadratic Equations

Solving by factorisation

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Introduction

A quadratic equation
is an equation of the form $ax^2 + bx + c = 0$ ($a \neq 0$).

Steps to solve by factorisation

1. Write the equation in the form $ax^2 + bx + c = 0$.
2. Factorise the left-hand side.
3. Set each factor equal to zero and solve for x.

EXAMPLE 1

$x^2 - 5x + 6 = 0$

$(x - 2)(x - 3) = 0$

$x = \mathbf{2}$ or $x = \mathbf{3}$

EXAMPLE 2

$2x^2 + 7x - 15 = 0 \rightarrow (2x - 3)(x + 5) = 0 \rightarrow x = \frac{3}{2}$ or $x = \mathbf{-5}$

Practice

(1) $x^2 + 2x - 24 = 0$

(2) $x^2 - 11x + 28 = 0$

(3) $3x^2 - 12x = 0$